Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EX5 – Elgamal Cryptography

Code:

Alice (Server):

```
import random

import socket


q = int(input("Enter the value of \'q\': "))

alpha = int(input("Enter a primitive root of \'q\': "))


X_a = random.randint(1, q-2)

Y_a = pow(alpha, X_a, q)

print(f"\nThe public key is: [{q}, {alpha}, {Y_a}].")


server = socket.socket()

server.bind(("127.0.0.1", 5678))

server.listen(1)

print("\nAlice has set up the server and is waiting for Bob.")

conn, addr = server.accept()

print(f"Bob has connected from {addr}.")

conn.send(str(q).encode().ljust(1024, b'\0'))

conn.send(str(alpha).encode().ljust(1024, b'\0'))

conn.send(str(Y_a).encode().ljust(1024, b'\0'))

print("The public key has been sent to Bob.")


C1 = int(conn.recv(1024).decode().rstrip('\0'))
```

```
C2 = int(conn.recv(1024).decode().rstrip('\0'))
print(f"Received ciphertext is: [{C1}, {C2}].")

K = pow(C1, X_a, q)
M = (C2 * pow(K, -1, q)) % q
print(f"\nThe message is: {M}.")

conn.close()
server.close()
```

Bob (Client):

```
import socket
import random

s = socket.socket()
s.connect(("127.0.0.1", 5678))
print("Bob has been initialized and connected to Alice.")
q = int(s.recv(1024).decode().rstrip('\0'))
alpha = int(s.recv(1024).decode().rstrip('\0'))
Y_a = int(s.recv(1024).decode().rstrip('\0'))
print(f"\nReceived public key is: [{q}, {alpha}, {Y_a}].")

M = int(input("\nEnter a plaintext less than the value of q: "))
k = random.randint(1, q-1)
K = pow(Y_a, k, q)
C1 = pow(alpha, k, q)
C2 = (K * M) % q
print(f"The ciphertext is: [{C1}, {C2}].")

s.send(str(C1).encode().ljust(1024, b'\0'))
s.send(str(C2).encode().ljust(1024, b'\0'))
```

```
print("The ciphertext has been sent to Alice.")

s.close()
```

Output:

Alice (Server_:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\Elgamal> python elg_a.py
Enter the value of 'q': 13
Enter a primitive root of 'q': 2

The public key is: [13, 2, 5].

Alice has set up the server and is waiting for Bob.
Bob has connected from ('127.0.0.1', 52484).
The public key has been sent to Bob.
Received ciphertext is: [5, 3].

The message is: 11.
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\Elgamal>
```

Bob (Client):

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\Elgamal> python elg_b.py
Bob has been initialized and connected to Alice.

Received public key is: [13, 2, 5].

Enter a plaintext less than the value of q: 11
The ciphertext is: [5, 3].
The ciphertext has been sent to Alice.
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-8\Elgamal>
```